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RESEARCH-ARTICLE

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Published: 22 August 2025

[Citation in BibTeX format](#)

ICDA 2025: 2025 2nd International
Conference on Intelligent Computing
and Data Analysis
August 22 - 24, 2025
Zhengzhou, China

Study on Key Factors Affecting the Impact of Online Gaming Experience on Occupational Recovery Experience: Insights from Moderating Analysis and Machine Learning

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Abstract

Objective: Based on the Conservation of Resources (COR) theory, this study systematically investigates the impact of online gaming experiences (*challenge, control, cooperation, and reward*) on occupational recovery experience, focusing on the moderating role of immersive experience.

Methods: A total of 134 occupationally stressed employees (mean age ~31, regular gamers) completed daily diaries for 14 days ($N = 1,876$ observations), reporting the online gaming experience scale (OGES), recovery experience questionnaire (REQ), and immersion experience scale (IES). Multilevel Structural Equation Modeling (MSEM) was used to test direct and interaction effects, supplemented by Random Forest Regression (RFR) for variable importance.

Results: Gaming dimensions significantly enhanced recovery experiences ($\gamma = 0.500, p < .001$), supporting COR-based resource accumulation. Immersive experience exhibited a significant positive moderating effect (Interaction $\gamma = 0.202, p < .05$), amplifying the restorative benefits of the game. RFR confirmed the GE*IMM interaction as the strongest predictor, refuting the hypothesized inverted-U-shaped effect.

Conclusions: Specific online gaming experiences function as effective recovery tools via COR mechanisms, with immersive engagement enhancing benefits. The findings guide restorative game design, employer well-being interventions, and mindful employee gaming to promote sustained workplace health.

CCS Concepts

• Applied computing - Law, social and behavioral sciences - Psychology;

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ACM ISBN 979-8-4007-2020-8/2025/08

<https://doi.org/10.1145/3772726.3772854>

Keywords

Online Gaming, Gaming Experience, Recovery Experience, Immersive experience

ACM Reference Format:

Zhenyu Tu and Zheng Huang. 2025. Study on Key Factors Affecting the Impact of Online Gaming Experience on Occupational Recovery Experience: Insights from Moderating Analysis and Machine Learning. In *2025 2nd International Conference on Intelligent Computing and Data Analysis (ICDA 2025)*, August 22–24, 2025, Zhengzhou, China. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3772726.3772854>

1 Introduction

Escalating occupational competition has intensified work-related stress, tension, and stress responses globally, establishing **recovery experiences (RE)** as critical for maintaining workforce well-being and productivity. Concurrently, online gaming has evolved into a prevalent leisure activity [1]. Research confirms the role of gaming in mitigating stress: 89% of players report using video games as a stress-relief tool, [2] while systematic reviews identify stress reduction as a core motivation for gameplay [3]. Empirical studies indicate that gaming facilitates psychological recovery through emotional regulation [4] and detachment [5], particularly for occupational groups. However, research systematically examining the mechanisms linking *multidimensional* gaming experiences to recovery remains limited. Prior work has disproportionately focused on gaming addiction or isolated social dynamics, neglecting integrated analyses of gaming dimensions, psychological states and recovery pathways.

This study adopts the **Conservation of Resources (COR) theory**, which posits that individuals accumulate psychological (e.g., self-efficacy and perceived control) and social resources to counter stressors. We propose four gaming dimensions—**challenge, control, cooperation, and reward**—as catalysts for resource accumulation, directly enhancing recovery experiences (psychological detachment, relaxation, mastery, and perceived control). Crucially, the psychological state of **immersive experience** (characterized by deep cognitive absorption during gameplay) may moderate this relationship. While moderate immersion may optimize engagement and reinforce benefits, excessive immersion can trigger time distortion and unintended resource depletion, potentially reversing recovery benefits, thereby suggesting a moderating effect.

This study pioneers a COR-based framework that examines how core online gaming dimensions facilitate psychological recovery through resource accumulation. We hypothesized that immersive experience exhibits a nonlinear moderating effect, enhancing recovery at moderate levels but diminishing it when it is excessive. To test this, 134 stressed professionals completed 14-day diaries (totaling 1,876 observations), reporting daily gaming states, immersion, and recovery. Random Forest Regression (RFR) analysis quantified moderation effects via Mean Decrease in Mean Squared Error (MD-MSE) and bootstrap robustness. The findings redefine the occupational health potential of gaming by establishing immersion thresholds relevant to game designers (optimizing restorative experiences), employers (supporting structured recovery interventions), and employees (promoting mindful engagement).

2 METHODS

2.1 Participants

We recruited 134 employees (aged 22–45 years; ≥ 3 years' work experience) via organizations. The inclusion criteria were as follows: (1) ≥ 1 year of regular gaming (≥ 2 sessions/week) and (2) occupational stress (COSS index > 1 , which assesses work demand/control imbalance). The Core Occupational Stress Scale (COSS) consists of 13 items, including five items for work demand and five items for control. It uses a Likert 5-point assignment setting with a value range of 1–5. The ratio of work demand/control is an occupational stress index. If its value is greater than 1, it is considered that the survey subject is in a state of occupational stress. Participants with psychological disorders, night-shift workers, or gaming addiction (IAT score ≥ 50) were excluded from the study.

2.2 Procedure & Measures

A 14-day diary study yielded 1876 valid responses. After baseline screening (demographics, stress, addiction), participants reported daily the following:

Recovery Experience Questionnaire (REQ) [6, 7]: 14 items measuring psychological detachment ($\alpha = 0.82–0.89$), relaxation ($\alpha = 0.73–0.86$), mastery ($\alpha = 0.82–0.88$), and control ($\alpha = 0.77–0.82$).

Online Gaming Experience Scale (OGES) [8]: Focused dimensions: challenge, control, cooperation, reward ($\alpha = 0.73–0.78$). The OGES is divided into 11 dimensions with a total of 44 questions. The 11 dimensions are challenge, fantasy, curiosity, control, role-playing, competition, cooperation, recognition, belongingness, responsibility, and rewards. All dimensions were measured using the Likert five-point scale

Immersion Experience Scale (IES) [9–11]: Five items assessing immersion intensity (e.g., "Felt time passed quickly"; 5-point Likert scale).

2.3 Statistical Analyses

Multilevel SEM (MSEM) with group-mean centering (within-person) and grand-mean centering (between-person) was used to test the direct effect of gaming experience dimensions on recovery experience and cross-level moderation of immersion experience $\times$ gaming experience interaction.

Structural equation of within-Person Moderation Model:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + \beta_2 W_{ij} + \beta_3 (X_{ij} \times W_{ij}) + e_{ij} \quad (1)$$

Here:

Y_{ij} : Recovery experience for person i at time j X_{ij} : Gaming experience (within-person centered)

W_{ij} : Immerse experience (within-person centered)

$X_{ij} \times W_{ij}$: Interaction term

e_{ij} : Within-person residual

Structural equation of between-Person Moderation Model:

$$Y_i = \gamma_0 + \gamma_1 X_i + \gamma_2 W_i + \gamma_3 (X_i \times W_i) + \sum_{k=4}^9 \beta_k C_{ki} + u_i \quad (2)$$

Here:

Y_i : Mean recovery experience for person i X_i : Between-person gaming experience (grand-mean centered)

W_i : Mean immerse experience (grand-mean centered)

$X_i \times W_i$: Cross-level interaction

C_{ki} : Covariates:

$C4i$: Gender

$C5i$: Age group

$C6i$: Highest education

$C7i$: Work tenure

$C8i$: Marital status

$C9i$: Gaming duration

u_i : Between-person residual

Random Forest Regression (RFR) was used to quantify moderation via MD-MSE, assess variable importance for recovery outcomes, and validate model robustness with bootstrap aggregation (ntrees = 500).

All statistical analyses were performed using R v4.4.3.

3 Results

3.1 Statistical descriptions of each variable

The study comprised 134 valid participants, with males representing 46.3% ($n = 62$) and females representing 53.7% ($n = 72$). The age distribution showed that 58.2% ($n = 78$) were 21–30 years old, 37.3% ($n = 50$) were 31–40 years old, and 3.7% ($n = 5$) were 41–50 years old, while one participant (0.7%) was under 20. Occupational backgrounds included 60.4% ($n = 81$) from private enterprises, 15.7% ($n = 21$) from state-owned enterprises, 11.9% ($n = 16$) from public institutions, 7.5% ($n = 10$) from foreign-funded enterprises, and 4.5% ($n = 6$) from government agencies. Education levels were dominated by associate degrees (71.6%, $n = 96$), followed by master's degrees (14.2%, $n = 19$) and vocational education (4.5%, $n = 6$). Work experience ranged primarily between 3–10 years (51.5%, $n = 69$), with 27.6% ($n = 37$) having 1–3 years, 17.2% ($n = 23$) 10–20 years, and 3.7% ($n = 5$) having less than one year. Most participants were married (63.4%, $n = 85$), while 36.6% ($n = 49$) were unmarried, and the average gaming frequency was 2.067 ± 1.007 sessions per week.

3.2 Variable Distribution Characteristics

Within-person variables exhibited a normal distribution, with a maximum absolute skewness of 1.253 and a maximum absolute kurtosis of 0.897, both below the threshold of 2, satisfying the normality assumptions for subsequent analyses. Between-person

variables also demonstrated normal distribution, showing a maximum absolute skewness of 1.026 and a maximum absolute kurtosis of 2.279, both under the critical value of 3, meeting the normality requirements for further statistical testing. The minimum, maximum, mean, and standard deviation values for all measurement items at both levels are presented in Table 1.

The between-subject variables and their measurement items are detailed in Table 2, presenting minimum values, maximum values, means, and standard deviations. The maximum absolute values for skewness and kurtosis were 1.026 and 2.279, respectively, both of which were below the critical value of 3. These results confirmed the normal distribution of the data, thereby meeting the parametric assumptions for subsequent statistical analyses.

3.3 correlation analysis

For within-person effects, significant positive correlations emerged between gaming experience and recovery experience ($r = 0.673, p < .01$), gaming experience and immerse experience ($r = 0.442, p < 0.01$), and immerse experience and recovery experience ($r = 0.541, p < 0.01$). For between-person effects, significant positive correlations were observed between gaming experience and recovery experience ($r = 0.456, p < .01$), gaming experience and immersive experience ($r = 0.307, p < 0.01$), immersive experience and recovery experience ($r = 0.329, p < 0.01$). For more information, see Table 3.

3.4 Moderating analysis

Multilevel analysis revealed a significant positive effect of gaming experience on recovery experience ($\gamma = 0.500, p < 0.001$). Immersive experience as a moderator also showed significant positive effects on the recovery experience ($\gamma = 0.237, p < 0.001$). Critically, the interaction term (gaming experience $\times$ immersive experience) demonstrated significant positive effects ($\gamma = 0.202, p < 0.05$), confirming immersive experience as a significant positive moderator between gaming experience and recovery outcomes. For more details, see Table 4.

Furthermore, the positive impact of gaming experience on recovery experience was significantly more pronounced in high immersion states than in low immersion states. This pattern indicates the progressive amplification of the restorative effects of gaming as the immersive experience intensifies (see Figure 1).

3.5 Random Forest analysis

The random forest algorithm was used to further explore and verify the predictive capability of GE and IMM for RE, particularly at the item level. This step aimed not only to build a predictive model but also to identify the most important predictors, providing a basis for future intervention strategies. We used the total scores and factor scores of the OGES and IES scales as independent variables to predict the total score of the REQ. In our RFR analysis, we ensured a consistent distribution of the dependent variable (GRE) between the training and test sets using R's createDataPartition() function from the caret package with $p = 0.75$ for a 75:25 split. The distribution consistency was verified by visually comparing the histograms generated by the hist() function before and after splitting. This approach maintains the original data distribution in both sets, thereby enhancing the generalizability of the results. A

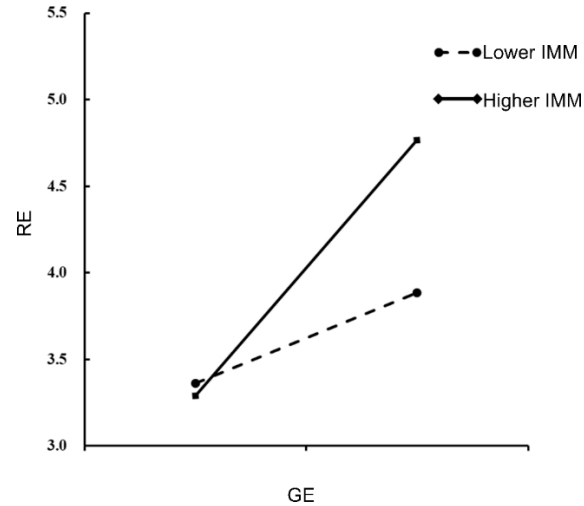


Figure 1: Immersion Experience Modulates the Relationship between Gaming Experience and Recovery Experience.

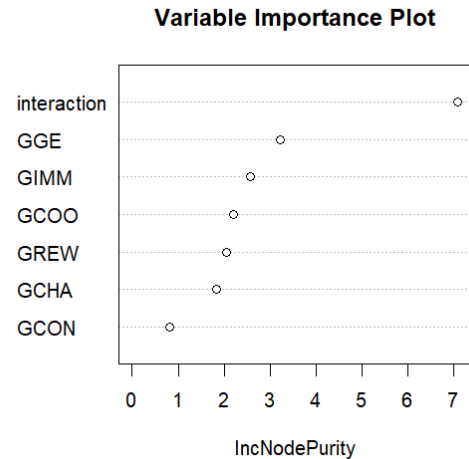


Figure 2: Variable importance plot of the random forest model.

training sample of 75% of the total sample was selected based on the distribution of the dependent variable, and the random forest method was used for training. The remaining 25% of the samples were used to validate the model. Figure 2 shows the importance of each variable in predicting dependent variables. The interaction factor of GE and IMM (GE*IMM) was the most important variable, indicating that the absence of the interaction factor would cause the greatest change in the entire model.

Interaction: the interacting factor of game experience and immerse experience; GGE: game experience; GIMM: immerse experience; GCOO: cooperation subscale of game experience; GREW: reward subscale of game experience; GCHA: challenge subscale of game experience; GCON: control subscale of game experience

Table 1: Characteristics of Within-person variables.

Variable/Item	N	Minimum	Maximum	Mean	SD	Skewness	Kurtosis
CON1	1876	1.000	5.000	3.328	1.230	-0.245	-0.897
CON2	1876	1.000	5.000	3.484	1.240	-0.389	-0.840
CON3	1876	1.000	5.000	3.778	1.220	-0.715	-0.531
CON	1876	1.000	5.000	3.530	1.014	-0.402	-0.728
CHA1	1876	1.000	5.000	3.417	1.238	-0.321	-0.877
CHA2	1876	1.000	5.000	3.799	1.217	-0.740	-0.491
CHA3	1876	1.000	5.000	3.704	1.231	-0.634	-0.639
CHA	1876	1.000	5.000	3.640	1.024	-0.535	-0.592
REW1	1876	1.000	5.000	3.602	1.244	-0.531	-0.764
REW2	1876	1.000	5.000	4.149	1.126	-1.173	0.369
REW3	1876	1.000	5.000	3.787	1.221	-0.726	-0.516
REW4	1876	1.000	5.000	3.738	1.227	-0.672	-0.598
REW5	1876	1.000	5.000	3.973	1.182	-0.936	-0.145
REW	1876	1.000	5.000	3.850	0.944	-0.763	-0.235
COO1	1876	1.000	5.000	3.956	1.186	-0.917	-0.191
COO2	1876	1.000	5.000	3.494	1.242	-0.401	-0.835
COO3	1876	1.000	5.000	3.722	1.231	-0.653	-0.623
COO4	1876	1.000	5.000	3.709	1.229	-0.636	-0.637
COO	1876	1.000	5.000	3.720	1.019	-0.581	-0.539
GE	1876	1.133	5.000	3.685	0.762	-0.470	-0.293
IMM1	1876	1.000	5.000	3.574	1.246	-0.500	-0.794
IMM2	1876	1.000	5.000	3.741	1.228	-0.676	-0.594
IMM3	1876	1.000	5.000	4.056	1.161	-1.041	0.053
IMM4	1876	1.000	5.000	3.652	1.241	-0.581	-0.712
IMM5	1876	1.000	5.000	4.107	1.147	-1.128	0.259
IMM	1876	1.000	5.000	3.826	0.948	-0.717	-0.242
SEP1	1876	1.000	5.000	3.696	1.232	-0.626	-0.651
SEP2	1876	1.000	5.000	3.869	1.200	-0.827	-0.328
SEP3	1876	1.000	5.000	4.195	1.103	-1.253	0.607
SEP	1876	1.000	5.000	3.920	0.975	-0.836	-0.120
REL1	1876	1.000	5.000	3.380	1.233	-0.288	-0.881
REL2	1876	1.000	5.000	3.757	1.223	-0.693	-0.567
REL3	1876	1.000	5.000	3.663	1.236	-0.591	-0.698
REL	1876	1.000	5.000	3.600	1.032	-0.468	-0.655
MAS1	1876	1.000	5.000	3.517	1.242	-0.456	-0.822
MAS2	1876	1.000	5.000	3.953	1.187	-0.913	-0.189
MAS3	1876	1.000	5.000	3.665	1.235	-0.592	-0.695
MAS4	1876	1.000	5.000	3.626	1.244	-0.555	-0.746
MAS	1876	1.000	5.000	3.690	0.988	-0.604	-0.491
CTR1	1876	1.000	5.000	4.094	1.151	-1.103	0.190
CTR2	1876	1.000	5.000	3.616	1.244	-0.547	-0.748
CTR3	1876	1.000	5.000	3.852	1.208	-0.810	-0.367
CTR4	1876	1.000	5.000	3.838	1.208	-0.795	-0.389
CTR	1876	1.000	5.000	3.850	0.991	-0.772	-0.249
RE	1876	1.063	5.000	3.765	0.757	-0.649	0.117

Notes: CON: control dimension in game experience; CHA: challenge dimension in game experience; REW: reward dimension in game experience; COO: cooperation dimension in game experience; GE: total score of game experience; IMM: immerse experience; SEP: Psychological Detachment dimension in recovery experience; REL: Relaxation dimension in recovery experience; MAS: Mastery Experiences dimension in recovery experience; CTR: control dimension in recovery experience; RE: total score of recovery experience

Table 2: Characteristics of Inter-individual variables.

Variable	N	Minimum	Maximum	Mean	SD	Skewness	Kurtosis
CON	134	2.024	4.643	3.530	0.568	-0.398	-0.306
CHA	134	2.381	4.738	3.640	0.576	-0.274	-0.761
REW	134	2.243	4.814	3.850	0.540	-0.700	0.022
COO	134	2.125	4.768	3.720	0.568	-0.495	-0.174
GE	134	2.560	4.629	3.685	0.478	-0.388	-0.515
IMM	134	1.643	5.000	3.826	0.523	-1.026	2.279
SEP	134	2.262	4.929	3.920	0.476	-0.816	0.834
REL	134	2.310	4.738	3.600	0.512	-0.227	-0.354
MAS	134	2.500	4.732	3.690	0.507	-0.258	-0.592
CTR	134	2.661	4.821	3.850	0.462	-0.473	-0.084
RE	134	2.548	4.805	3.765	0.422	-0.445	0.244

Notes: CON: control dimension in game experience; CHA: challenge dimension in game experience; REW: reward dimension in game experience; COO: cooperation dimension in game experience; GE: total score of game experience; IMM: immerse experience; SEP: Psychological Detachment dimension in recovery experience; REL: Relaxation dimension in recovery experience; MAS: Mastery Experiences dimension in recovery experience; CTR: control dimension in recovery experience; RE: total score of recovery experience

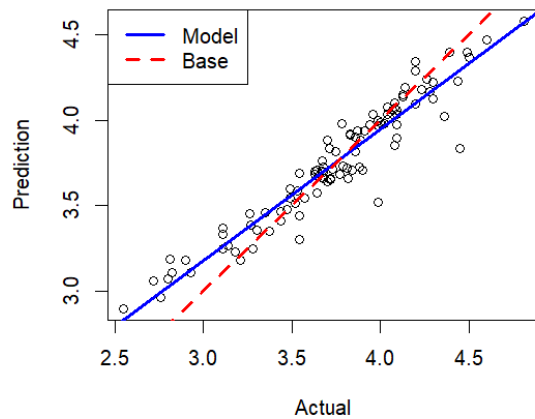


Figure 3: Comparison between the real value and the model predicted value in the training set.

Next, we tested the predictive performance of the model. In the training set, the Root Mean Squared Error (RMSE) was 0.152, R-squared (R^2) was 0.906, and the Mean Absolute Error (MAE) was 0.108. In the test set, the RMSE, R-squared (R^2), and MAE values were 0.275, 0.397, and 0.200, respectively. Figures 3 and Figure 4 compare the actual values and the predicted values in the training and test sets.

4 Discussion

This study pioneers a Conservation of Resources (COR) framework to elucidate how distinct online gaming experiences facilitate psychological recovery in occupationally stressed employees. Utilizing rigorous, ecologically valid methods (daily diary tracking over 14

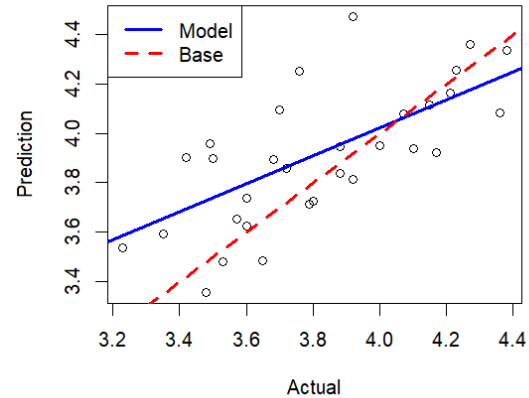


Figure 4: Comparison of the true values in the test set with the model predicted values.

days) and sophisticated analytic approaches (Multilevel SEM, Random Forest Regression), our findings significantly advance the understanding of the restorative potential of gaming beyond anecdotal reports, revealing specific mechanisms and boundary conditions.

Research findings indicate that gaming experiences (challenge, control, cooperation, reward) have a significant positive effect on recovery experiences, which aligns with the perspective of COR theory. Specifically, cooperation and reward fulfill individuals' needs for belonging and achievement, consequently enhancing recovery experiences [12, 13]. According to the RFR, cooperation and reward are the two most important factors among the four subfactors of GE. For challenge, moderate in-game challenges can stimulate an individual's sense of achievement and sense of control, promoting the replenishment of psychological resources (specifically, such as

Table 3: Correlation analysis

		01	02	03	04	05	06	07	08	09	10	11
Between-person variable (N=134)	01	1										
	CON											
	02	.548**	1									
	CHA											
	03	.663**	.579**	1								
	REW											
	04	.637**	.698**	.648**	1							
	COO											
	05 GE	.838**	.835**	.846**	.879**	1						
	06	.395**	.310**	.486**	.316**	.442**	1					
	IMM											
Within-person variable (N=1876)	07 SEP	.498**	.540**	.563**	.486**	.614**	.447**	1				
	08	.421**	.488**	.480**	.510**	.559**	.470**	.648**	1			
	REL											
	09	.385**	.494**	.444**	.522**	.543**	.503**	.632**	.628**	1		
	MAS											
	10	.482**	.532**	.522**	.541**	.612**	.445**	.699**	.718**	.635**	1	
	CTR											
	11 RE	.516**	.594**	.580**	.597**	.673**	.541**	.859**	.871**	.843**	.879**	1
	01	1										
	CON											
	02	.419**	1									
	CHA											
	03	.465**	.434**	1								
	REW											
	04	.405**	.460**	.463**	1							
	COO											
	05 GE	.753**	.763**	.765**	.767**	1						
	06	.222**	.212**	.290**	.216**	.307**	1					
	IMM											
	07 SEP	.257**	.289**	.278**	.242**	.349**	.238**	1				
	08	.263**	.250**	.256**	.241**	.332**	.242**	.410**	1			
	REL											
	09	.232**	.288**	.258**	.261**	.341**	.287**	.439**	.449**	1		
	MAS											
	10	.282**	.277**	.299**	.255**	.365**	.233**	.426**	.448**	.446**	1	
	CTR											
	11 RE	.340**	.363**	.359**	.329**	.456**	.329**	.744**	.766**	.767**	.762**	1

Notes: CON: control dimension in game experience; CHA: challenge dimension in game experience; REW: reward dimension in game experience; COO: cooperation dimension in game experience; GE: total score of game experience; IMM: immerse experience; SEP: Psychological Detachment dimension in recovery experience; REL: Relaxation dimension in recovery experience; MAS: Mastery Experiences dimension in recovery experience; CTR: control dimension in recovery experience; RE: total score of recovery experience

psychological detachment and mastery experiences). This is consistent with previous research [14, 15], which found that optimally challenging activities facilitate recovery from work stress. Finally, as a control factor, the sense of autonomy and control within games enhances players' self-efficacy [16], thereby promoting relaxation and fostering a sense of control over leisure time.

Contrary to our hypothesis predicting an inverted U-shaped moderating effect, multilevel SEM revealed a consistent, positive moderating effect of immersive experience. Specifically, deeper

immersion during gameplay amplified the positive relationship between gaming dimensions and recovery, irrespective of immersion intensity levels captured in our sample. The RFR results further solidified the centrality of this interaction, identifying the Gaming Experience * Immersion Experience (GE*IMM) interaction as the most important predictor of recovery across both the training and test datasets. This challenges the initial theoretical concern that "excessive" flow inevitably depletes resources and reverses the benefits.

Table 4: Significance test of moderating effect.

	nindependent variable	Dependent Variable (Recovery experience)
Within-person	Intercept	4.042*** (0.195)
	GE	0.323*** (0.041)
	IMM	0.130*** (0.032)
	GE×IMM	0.009 (0.036)
between-person	GE	0.500*** (0.06)
	IMM	0.237*** (0.064)
	GE×IMM	0.202* (0.097)
	Gender	-0.041 (0.06)
	age	0.099 (0.057)
	education	-0.029 (0.036)
	years of working	-0.046 (0.048)
	marriage	-0.026 (0.062)
	Gaming time	-0.069** (0.025)

The prominence of the GE*IMM interaction term in the RFR analysis underscores a critical theoretical insight: gaming’s restorative power lies not merely in the *presence* of core dimensions or immersion *alone* but in their dynamic interaction. The *synergy* between actively engaging with meaningful challenges, exercising control, collaborating, and experiencing rewards, all while being cognitively absorbed, most effectively rebuilds depleted psychological resources. This finding helps reconcile previous fragmented research focusing on isolated elements (such as social aspects) by emphasizing the integrated nature of the gaming recovery experience.

Although this study has its merits, it has the following limitations. First, the findings are limited to younger (21-40yo), associate-degree Chinese private-sector employees. In the future, we can expand to diverse demographics, occupations (e.g., healthcare), and gaming habits (e.g., casual players). Second, while immersion moderated the relationship, individual differences may exist. Future research could explore whether factors such as game genre, motivation for play (e.g., escapism vs. achievement), or specific personality traits (e.g., trait absorption, impulsivity) moderate the immersion-recovery link differently. Third, the absence of a non-gaming control group or a robust comparison to alternative recovery activities makes it challenging to definitively attribute the observed recovery benefits solely to online gaming or to ascertain whether gaming is more

effective than other forms of leisure. Finally, the study’s reliance on a homogeneous sample (younger, associate-degree Chinese private-sector employees) significantly restricts the generalizability of the findings to broader populations, diverse occupations, or varying cultural contexts.

5 Conclusions

The challenge, control, cooperation, and reward dimensions of online gaming promote psychological recovery by accumulating resources (COR theory). Immersion amplifies this effect, enhancing the benefits of typical play. These findings inform restorative game design, employer intervention, and mindful employee engagement.

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